

Unsupervised Classification: the Clustering Task

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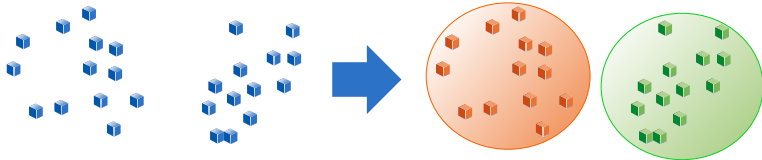


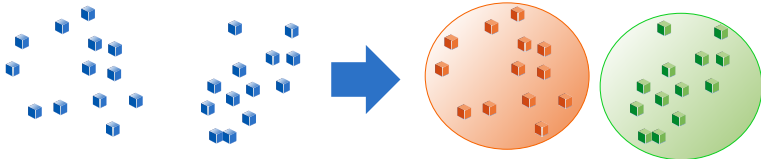
- 1 Introduction
- 2 Necessary Tools
- 3 Partitional Algorithms
- 4 Hierarchical Algorithms
- 5 How Many Clusters?
- 6 To Not Conclude



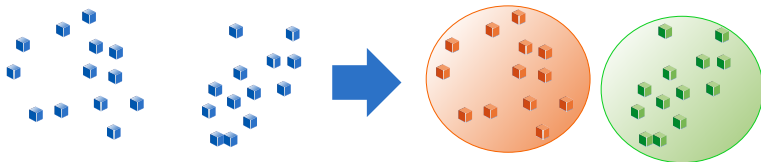
1 Introduction



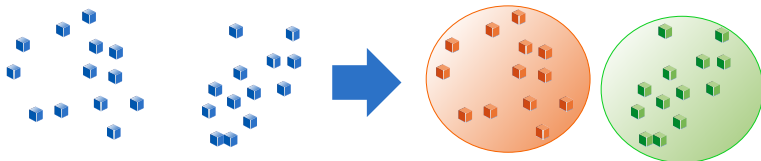




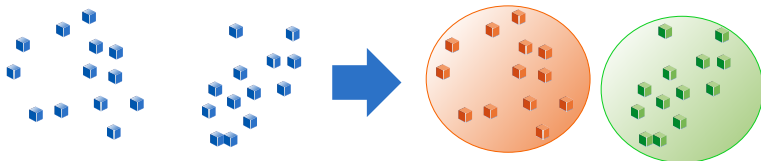
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- Similarity of data in one cluster (intra-cluster similarity) must be higher and similarity between clusters (inter-cluster similarity) must be lower.
- Then we need to measure the similarity/dissimilarity between objects and/or groups...

- Exploratory Data Analysis (typology) :



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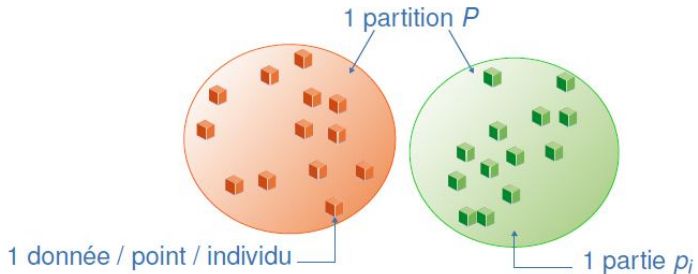


- Exploratory Data Analysis (typology) :
 - marketing : clients typology,
 - bio-computer science: genes grouping,
 - image : segmentation in homogeneous zones.
- Data Simplification :
 - information research : web pages grouping,
 - big data summary.



Definition - Partitioning

Divide a dataset into parts



Definition - **Partition** $P = \{p_1, p_2, \dots, p_K\}$ of a dataset D

P satisfies the following properties :

- $\cup_{i=1}^K p_i = D$
The parts of P contain all the data
- $p_i \cap p_j = \emptyset \forall i \neq j$
The parts are disjointed two by two
- $p_i \neq \emptyset$
All parts are non empty



What is the objective ?

Partitioning problem

Find the "best" partition... for a given objective.



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- There are many objective functions.
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- There are many objective functions.
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It is generally intended that the parties should be

- as far apart as possible from each other- (large distance between points not in the same parts).
- as homogeneous as possible - small distance between the points in the same part.

Definition - **Mincut** (minimize the similarity inter-clusters)

Let π the set of all partitions of a dataset D , let $i, j \in D$ We denote by $s_{i,j}$ the similarity between i and j The mincut function is defined as

$$\min_{P \in \pi} \frac{1}{2} \sum_{p \in P} \sum_{i \in p, j \notin p} s_{i,j}$$

This is equivalent to maximize the similarity intra-clusters.



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Definition - **RatioCut**

$$\frac{1}{2} \sum_{p \in P} \sum_{i \in p, j \notin p} \frac{s_{i,j}}{|\mathbf{p}|}$$

Definition - **NCut**

$$\frac{1}{2} \sum_{p \in P} \sum_{i \in p, j \notin p} \frac{s_{i,j}}{\text{vol}(\mathbf{p})}$$

Definition - Stirling number

$$S(n, K) = \frac{1}{K!} \sum_{j=1}^K (-1)^{K-j} \binom{K}{j} j^n$$

Corresponds to the number of K -partitions of n individuals.

Examples

K	S(30 K)
1	1
2	536 870 911
3	34 314 651 811 530
4	48 004 081 105 038 305
5	7 713 000 216 608 565 075
6	299 310 102 746 948 685 757
7	4 168 916 722 553 086 402 080
8	26 383 018 684 048 108 297 800
9	88 300 984 248 924 568 770 870
10	173 373 343 599 189 364 594 756

It is impossible to list all the solutions!



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2 Necessary Tools



Distance Measures

A distance measure d between two objects must fulfill the following criteria:

- 1 separation: $d(u, v) = 0 \Leftrightarrow u = v$,
- 2 symmetry: $d(u, v) = d(v, u)$,
- 3 triangle inequality: $d(u, v) \leq d(u, w) + d(w, v)$.



Ex. for $u = (u_1, \dots, u_n), v = (v_1, \dots, v_n) \in \mathbb{R}^n$,

- Manhattan (1-distance) $d_1(u, v) = \sum_{k=1}^n |u_k - v_k|$
- Euclidean (2-distance) $d_2(u, v) = \sqrt{\sum_{k=1}^n (u_k - v_k)^2}$
- Minkowski (p-distance) $d_p(u, v) = \sqrt[p]{\sum_{k=1}^n |u_k - v_k|^p}$
- Tchebychev (∞ -distance) $d_\infty(u, v) = \max_{1 \leq k \leq n} |u_k - v_k|$



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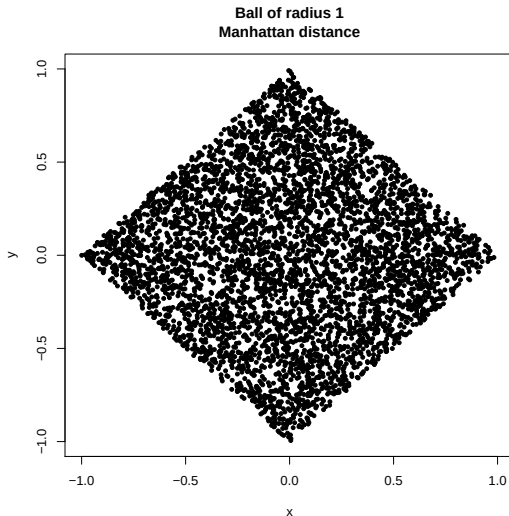
Cluster $\approx$ Ball:

A ball of center c and radius ϵ in a domain $\mathcal{D}$ is the following set:

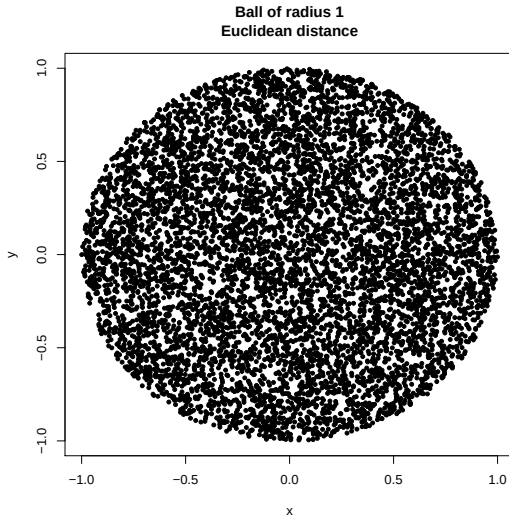
$$B(c, \epsilon) = \{x \in \mathcal{D} | d(c, x) \leq \epsilon\}$$



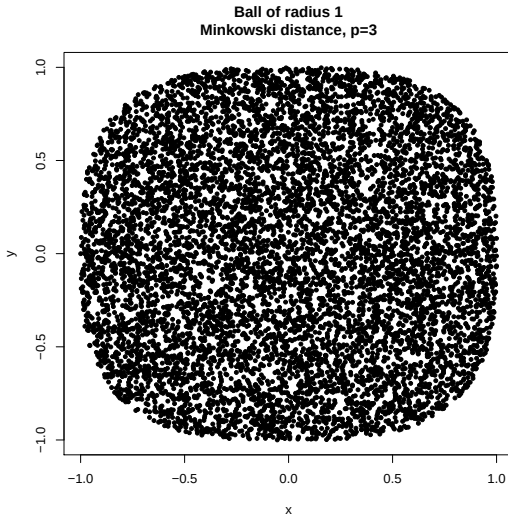
Manhattan (1-distance): $d_1(u, v) = \sum_{k=1}^n |u_k - v_k|$



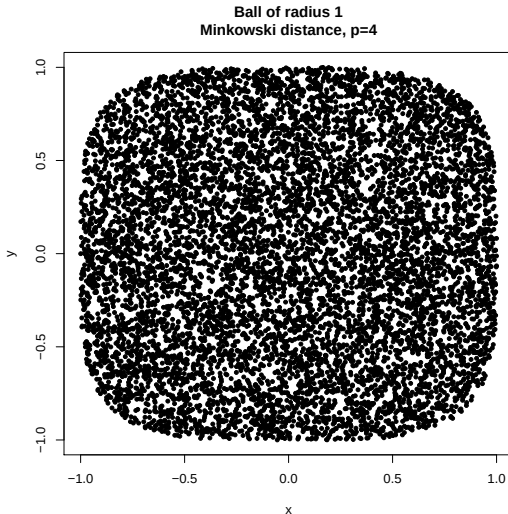
Euclidean (2-distance): $d_2(u, v) = \sqrt{\sum_{k=1}^n (u_k - v_k)^2}$



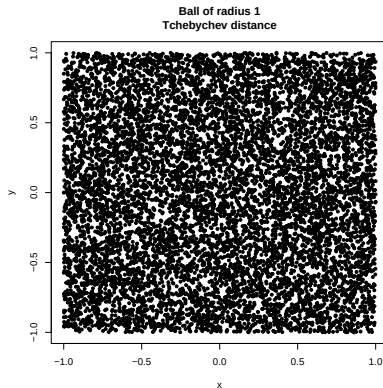
Minkowski (p-distance): $d_p(u, v) = \sqrt[p]{\sum_{k=1}^n |u_k - v_k|^p}$

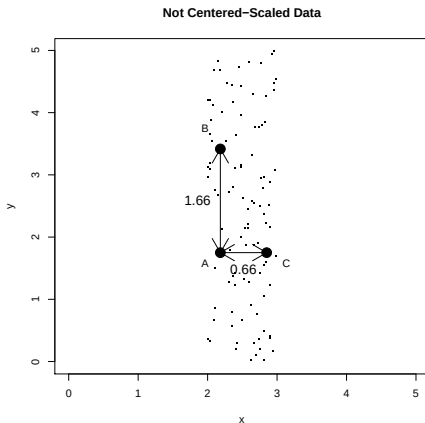


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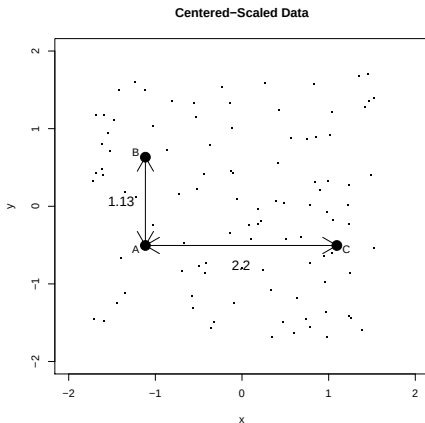


Tchebychev (∞ -distance): $d_p(u, v) = \max_{1 \leq k \leq n} |u_k - v_k|$





$$d(A, B) > d(A, C)$$



$$d(A, B) < d(A, C)$$

Distance Measures

A distance measure d between two objects must fulfill the following criteria:

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Similarity Measures

A similarity measure s must fulfill the following criteria:

- 1 $s(u, u) = k$, where k is a constant,
- 2 $s(u, v) = s(v, u)$,
- 3 $s(u, v) \leq s(u, u) = k$.



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Link Between Similarities and Distances

A distance measure d can be converted into a similarity measure using a strictly decreasing functions ϕ with some boundary conditions :

$$d = \phi(s) \text{ and } s = \phi^{-1}(d).$$

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Relocation type method

- Create an initial partition (usually randomly)
- Iteratively reallocate individuals between the K clusters : gradual improvement of the current solution.



3 Partitional Algorithms

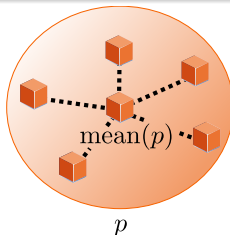
- k-means
- k-means++
- k-medoids



Let's start by defining the objective function of this algorithm.

Definition - Variability of a cluster p

$$var(p) = \sum_{e \in p} dist(mean(p), e)$$



Remark

Notion close to the variance but we don't divide by the size of p .

Definition - Weight of a partition P

$$diss(P) = \sum_{p \in P} var(p)$$

Objective function

Find a partition of minimize weight.



k-means algorithm

Starting: Randomly select k centroids (centers) $(c_1^{(0)} \dots, c_k^{(0)})$,

Finish $\leftarrow$ FALSE;

While : Finish = FALSE :

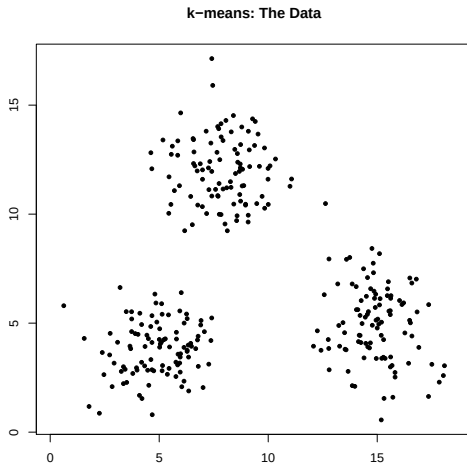
Assignment: observations go to the cluster with the closest centroid:

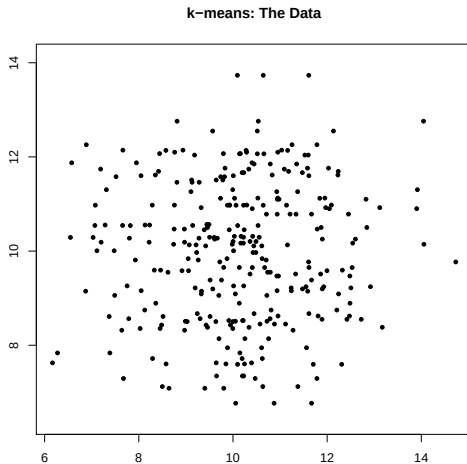
$$C_i^{(t+1)} = \left\{ x : d(x, c_i^{(t)}) \leq d(x, c_j^{(t)}) \forall j = 1, \dots, k \right\}$$

Update: Compute the new centroids $(c_1^{(t+1)}, \dots, c_k^{(t+1)})$:

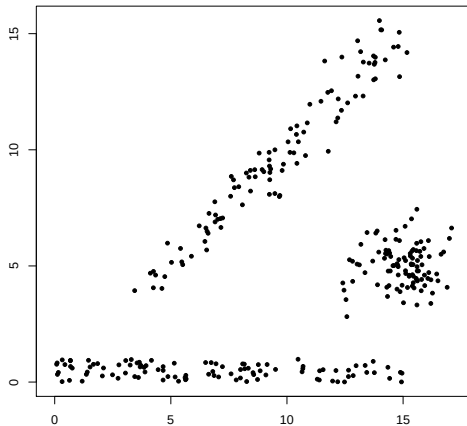
$$c_i^{(t+1)} = \frac{1}{|C_i^{(t)}|} \sum_{x \in C_i^{(t)}} x$$

Condition: If the centroids haven't moved : Finish $\leftarrow$ TRUE

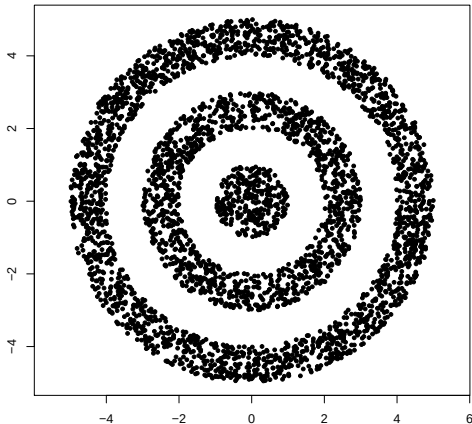




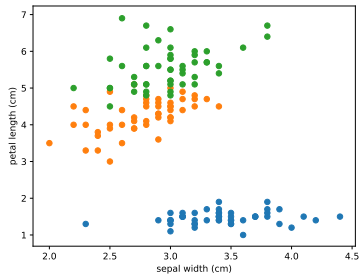
k-means: The Data



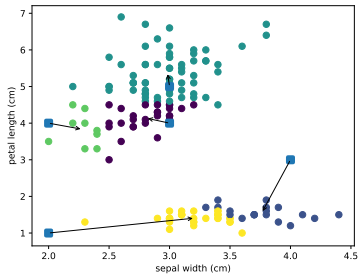
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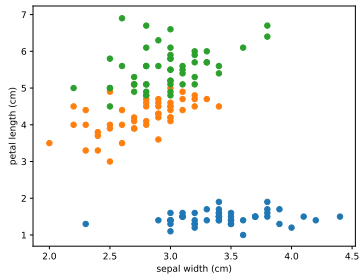
Real Partition of the iris



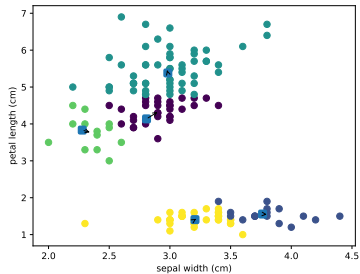
k-means - $k = 5$



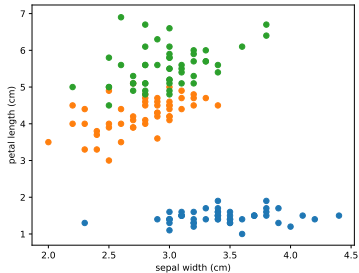
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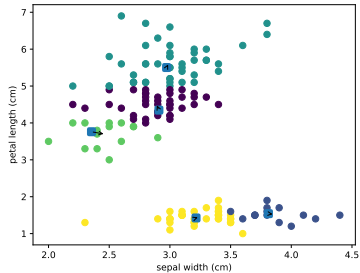
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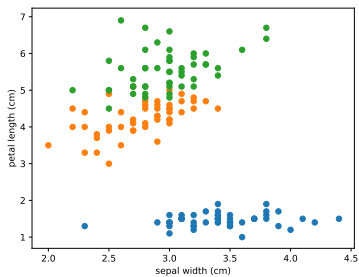
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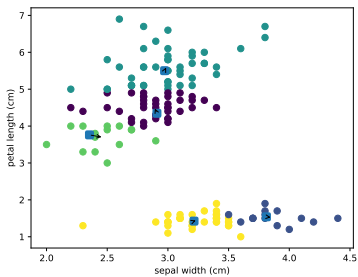
k-means - $k = 5$



Real Partition of the iris



k-means - $k = 5$



Whatever the parameters, the algorithm will always provide a solution. This does not mean that it is relevant.



- choice of k : an inappropriate choice leads too non significant results,
- spherical clusters: algorithms works better when spherical clusters are in data,
- instability: the random starting partition can leads to a local optimum for the criterion function.

Complexity: $O(m \times n \times k \times i)$ where m is the number of attributes, n the number of objects to cluster, k the number of clusters and i is the number of iterations required for convergence [FU15] .



- a k-means drawback : the random starting partition can leads to a local optimum for the criterion function.
- a paper : Arthur, D., Vassilvitskii, S. "k-means++: the advantages of careful seeding", 18th ACM-SIAM symposium on Discrete algorithms, 2007.



- 1 Take one center c_1 , chosen uniformly at random from data $\mathcal{X}$.
- 2 Take a new center c_i , choosing $x \in \mathcal{X}$ with probability

$$\frac{D(x)^2}{\sum_{x \in \mathcal{X}} D(x)^2}$$

where $D(x)$ denote the shortest distance from a data point to the closest center we have already chosen.

- 3 Repeat Step 2 until we have taken k centers altogether.
- 4 Proceed as with the standard k-means algorithm.

k-means++ is $O(\log k)$ competitive.



k-medoids or partitioning around medoids (PAM):

- 1 Initialize: greedily select k of the n data points as the medoids to minimize the cost.
- 2 Associate each data point to the closest medoid.
- 3 While the cost of the configuration decreases:
 - 1 For each medoid m , and for each non-medoid data point o :
 - 1 Consider the swap of m and o , and compute the cost change,
 - 2 If the cost change is the current best, remember this m and o combination
 - 2 Perform the best swap of m_{best} and o_{best} , if it decreases the cost function. Otherwise, the algorithm terminates.

A naive implementation is $O(n^2k^2)$, but can be reduced to $O(n^2)$, by splitting the cost change into three parts such that computations can be shared or avoided.



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4

Hierarchical Algorithms

- Agglomerative Hierarchical Algorithms
- Distances between 2 sets



Definition - Hierarchy of partitions $H = \{P_1, P_2, \dots, P_q\}$

Set of q partitions such that for all $i \neq j$ and $k > l \in \{1, \dots, n\}$

- $p_i \in P_k, p_j \in P_l$ implies
- $p_j \subset p_i$ ou $p_i \cap p_j = \emptyset$



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Hierarchical Algorithms

Starting: Put each individual alone in a cluster;

While : there is more than 1 cluster :
Merging the two closest parts.



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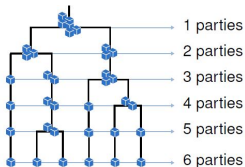
Hierarchical Algorithms

Starting: Put each individual alone in a cluster;

While : there is more than 1 cluster :
Merging the two closest parts.

We will have to define the distance between two sets





Definition - Dendogram

Graphical representation of a partition hierarchy



Single-link

- $dist(p_1, p_2) = \min_{i \in p_1, j \in p_2} dist(i, j)$
Smallest distance between a point of p_1 and a point of p_2 .

Complete-link

- $dist(p_1, p_2) = \max_{i \in p_1, j \in p_2} dist(i, j)$
Largest distance between a point of p_1 and a point of p_2 .

Average-link

- $dist(p_1, p_2) = mean_{i \in p_1, j \in p_2} dist(i, j)$
Average distance between a point of p_1 and a point of p_2 .

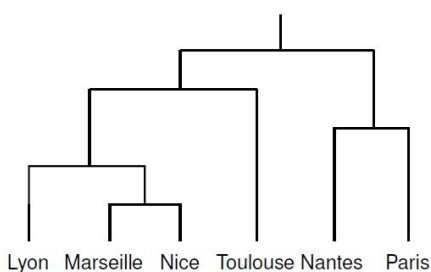
Complexity

$O(n^2)$ for the single linkage and $O(n^2 \log n)$ for complete and average linkages.

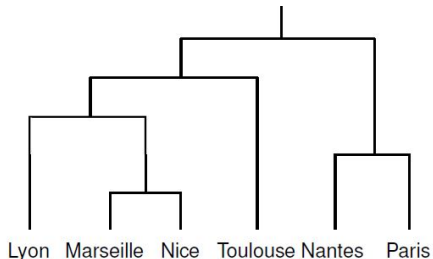
Distance between cities

	Lyon	Marseille	Nantes	Nice	Paris	Toulouse
Lyon	0	323	607	480	470	542
Marseille	323	0	911	197	793	412
Nantes	607	911	0	1068	393	579
Nice	480	197	1068	0	950	594
Paris	470	793	393	950	0	681
Toulouse	542	412	579	594	681	0





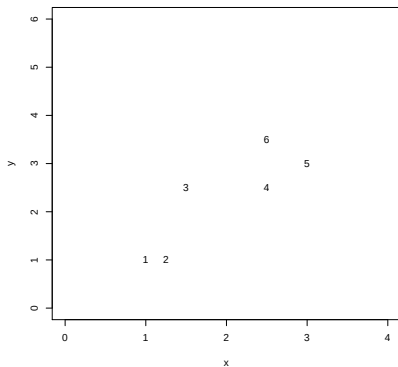
Single-link



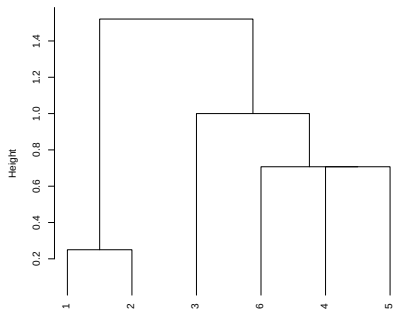
Complete-link



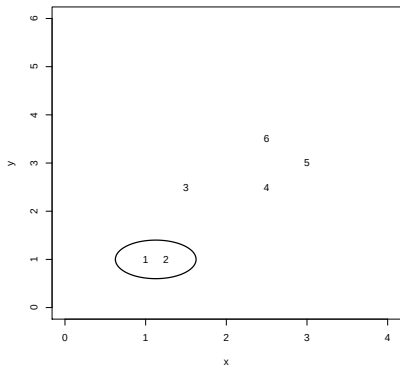
Hierarchical Clustering: The Data



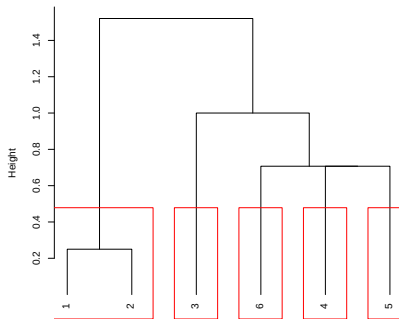
HClust, Single Link



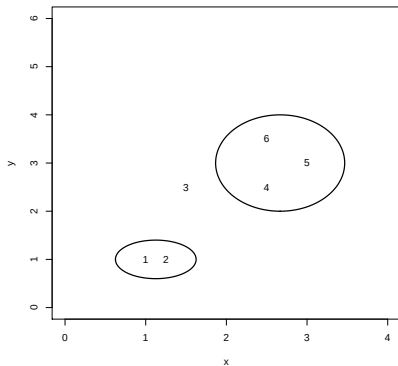
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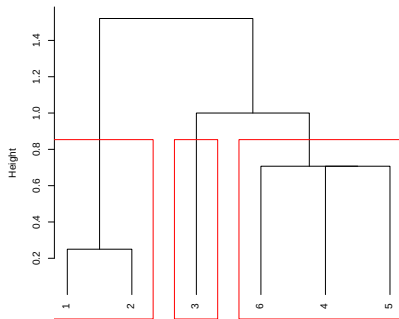
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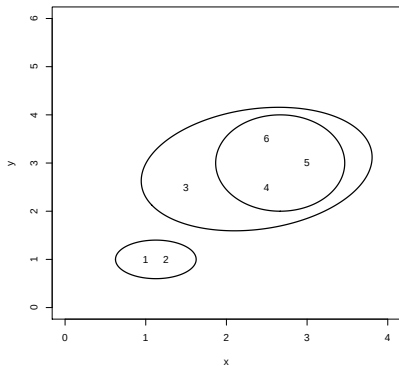
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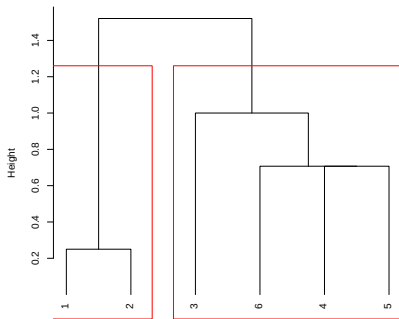
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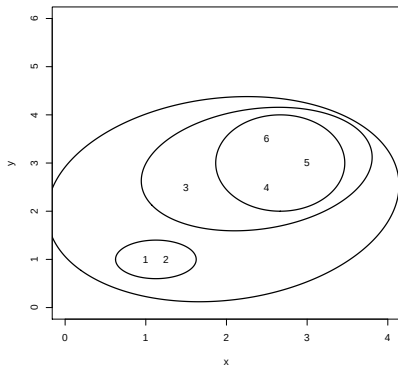
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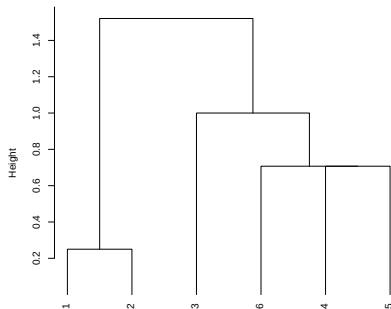
HClust, Single Link



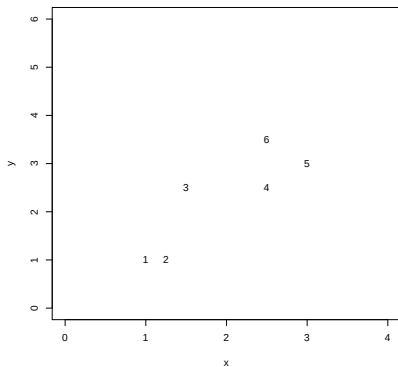
Hierarchical Clustering: The Data



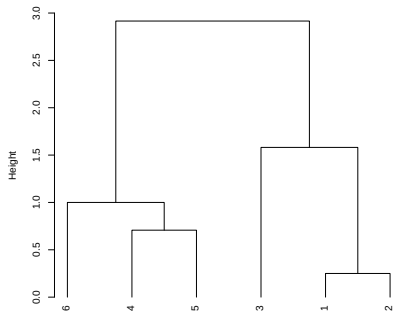
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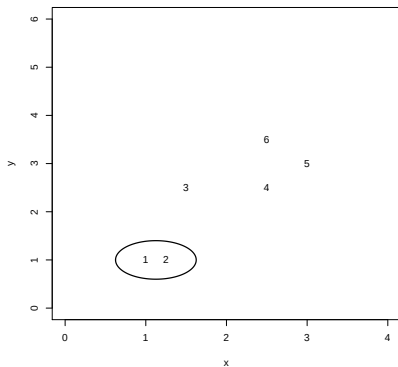
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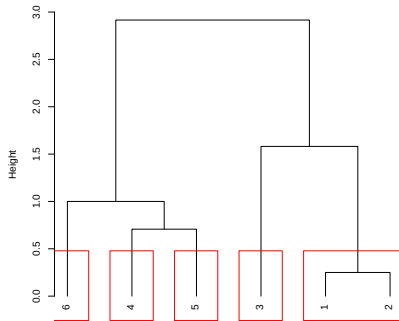
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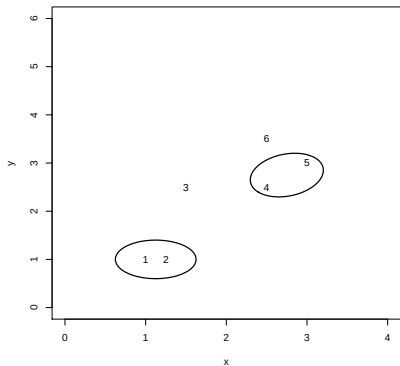
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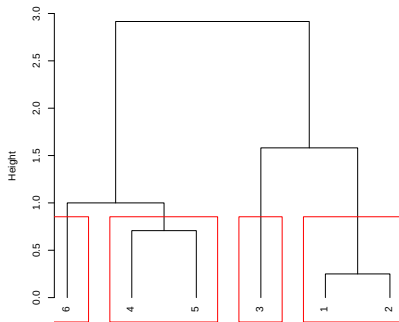
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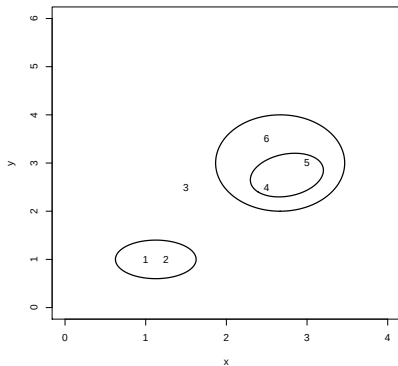
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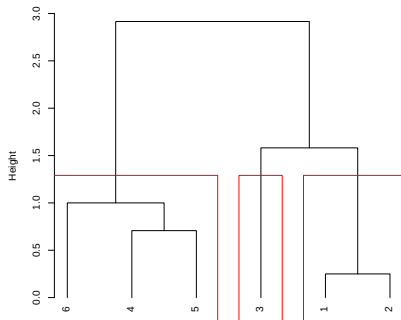
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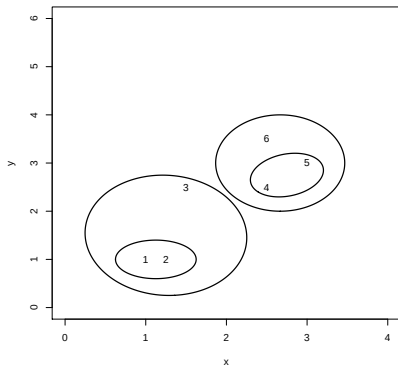
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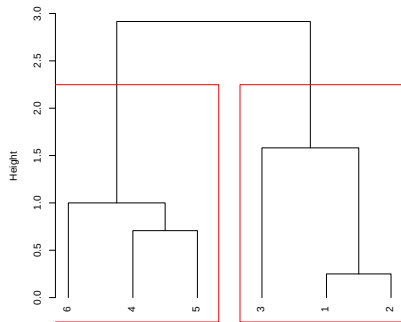
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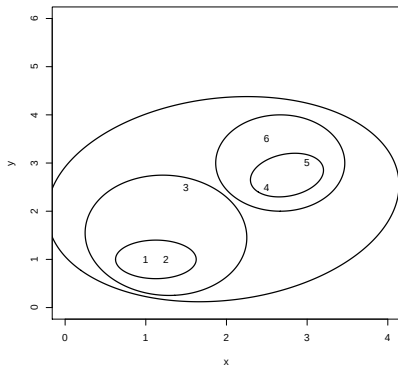
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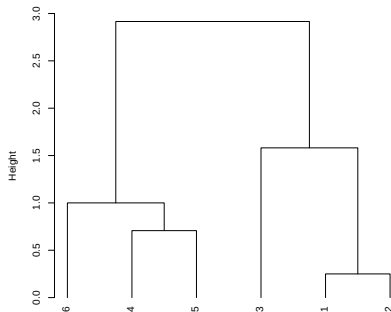
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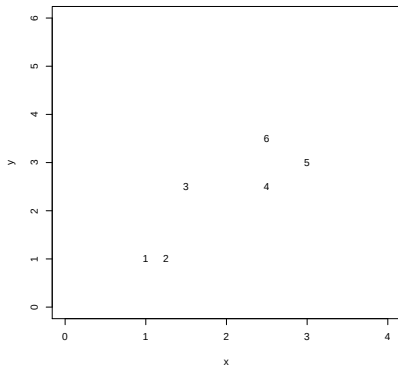
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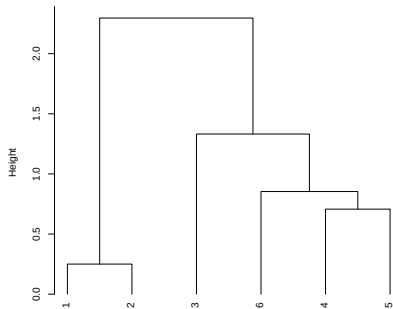
HClust, Complete Link



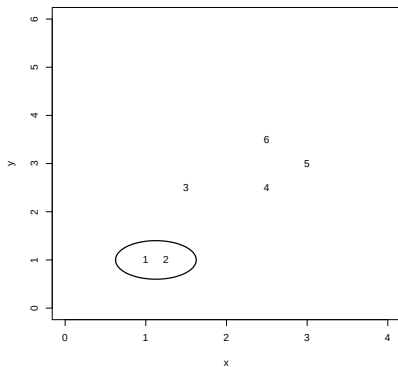
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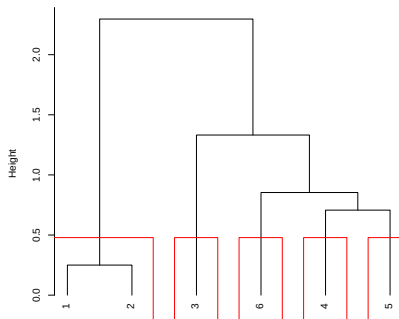
HClust, Average Link



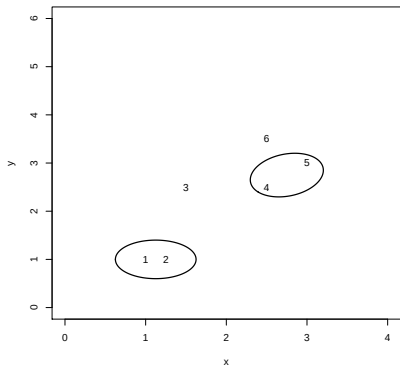
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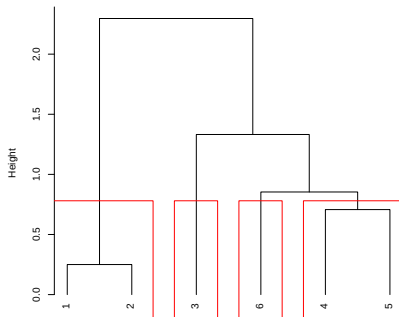
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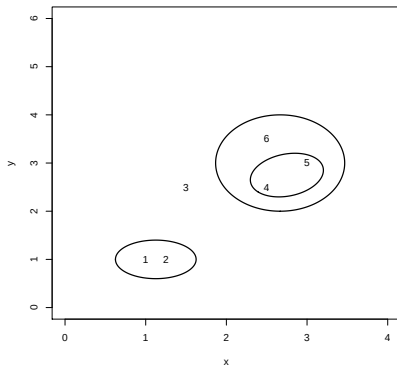
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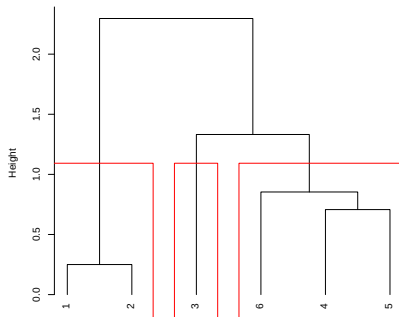
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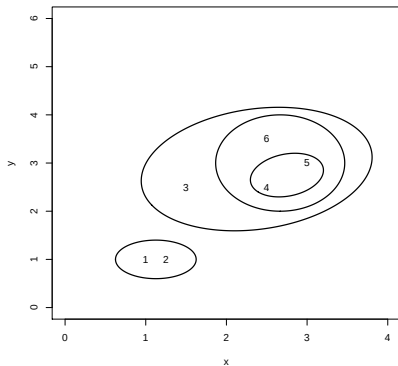
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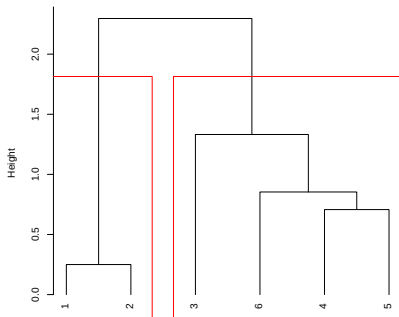
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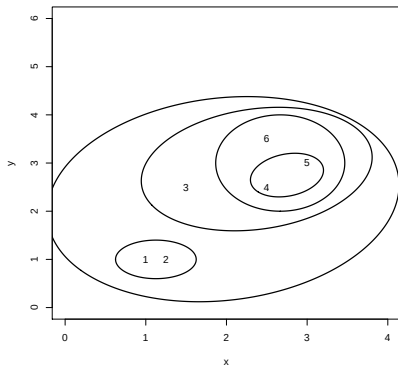
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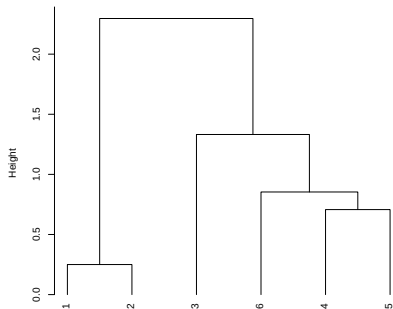
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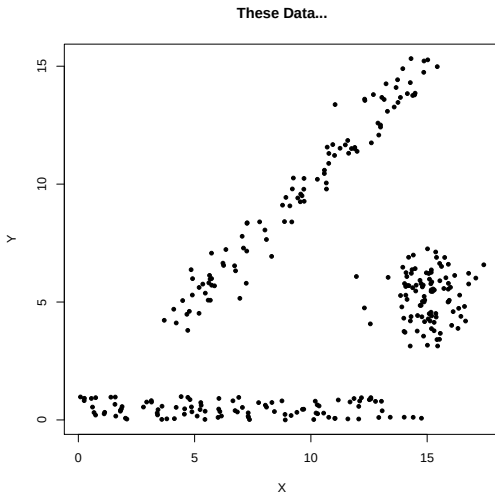


Complexity: $O(n^2)$ for the single linkage and $O(n^2 \log n)$ for complete and average linkages.

Solution:

Perform a simple k-means to summarize data, using a great value for k , but such the number of clusters is small enough to permits the hierarchical clustering.





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- 5 How Many Clusters?
 - The Elbow Method
 - The Silhouette Method
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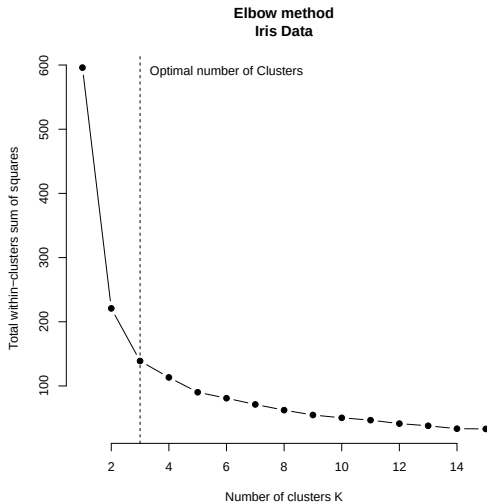
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- 3 Find the location of a bend (knee) in the plot, it's the suggested “optimal” k .





5 How Many Clusters?

- The Elbow Method
- **The Silhouette Method**
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- Assume the data have been clustered into k clusters.
- For each datum i , let $a(i)$ be the average dissimilarity of i with all other data within the same cluster.
- $a(i)$ as how well i is assigned to its cluster (the smaller the value, the better the assignment).
- Let $b(i)$ be the lowest average dissimilarity of i to any other cluster, of which i is not a member.
- The cluster with this lowest average dissimilarity is said to be the "neighbouring cluster" of i because it is the next best fit cluster for point i .

We now define a silhouette:

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Which can be also written as:

$$s(i) = \begin{cases} 1 - a(i)/b(i), & \text{if } a(i) < b(i) \\ 0, & \text{if } a(i) = b(i) \\ b(i)/a(i) - 1, & \text{if } a(i) > b(i) \end{cases}$$

From the above definition it is clear that

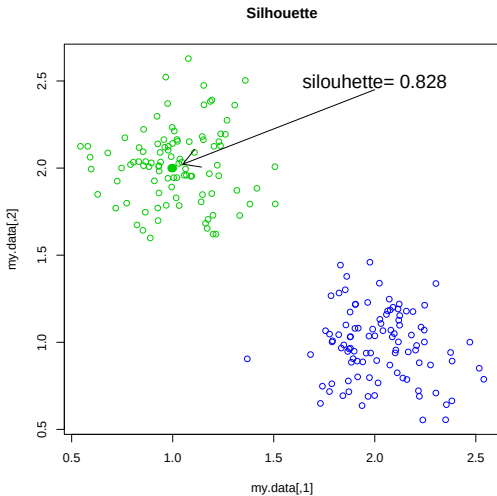
$$-1 \leq s(i) \leq 1$$

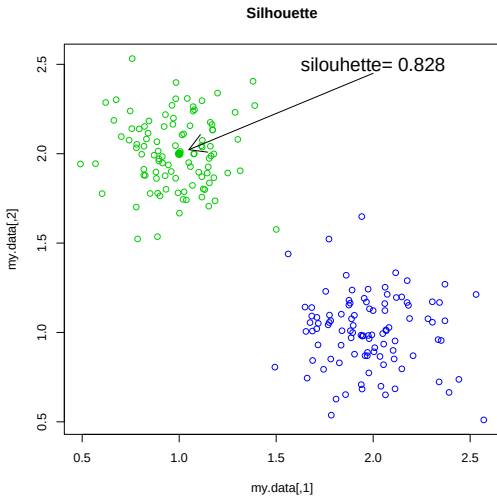


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- $s(i)$ is close to 1 when $a(i) \ll b(i)$,
- as $a(i)$ is a measure of how dissimilar i is to its own cluster, a small value means it is well matched,
- furthermore, a large $b(i)$ implies that i is badly matched to its neighbouring cluster,
- thus an $s(i)$ close to 1 means that the datum is appropriately clustered.
- If $s(i)$ is close to -1 , then i would be more appropriate if it was clustered in its neighbouring cluster.
- An $s(i)$ 0 means that the datum is on the border of two natural clusters.

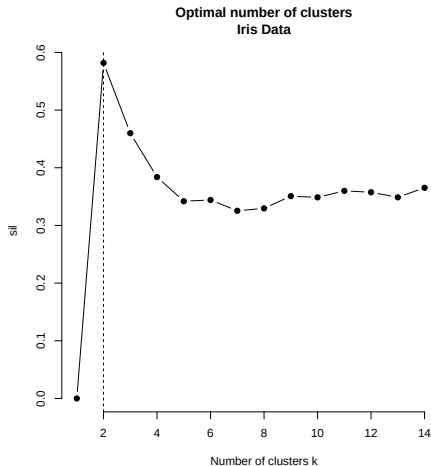






- The average $s(i)$ over all data of a cluster is a measure of how tightly grouped all the data in the cluster are.
- Thus the average $s(i)$ over all data of the entire dataset is a measure of how appropriately the data have been clustered.
- Average silhouette method computes the average silhouette of observations for different values of k .
- The optimal number of clusters k is the one that maximize the average silhouette over a range of possible values for k (Kaufman and Rousseeuw [1990]).





5 How Many Clusters?

- The Elbow Method
- The Silhouette Method
- **The Gap Statistic Method**
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- The gap statistic has been published by R. Tibshirani, G. Walther, and T. Hastie (Stanford University, 2001).
- The approach can be applied to any clustering method and is based on total within intracluster variation. If C_r is one of the k clusters of a partition:

$$D_r = \sum_{i,j \in C_r} d_{ij}$$

$$W_k = \sum_{r=1}^k \frac{1}{2n_r} D_r$$

where $n_r = |C_r|$.

- The gap statistic compares the total within intracluster variation for different values of k with their expected values under null reference distribution of the data, i.e. a distribution with no obvious clustering (uniform distribution).



- The reference dataset is generated using Monte Carlo simulations of the sampling process.
- That is, for each variable (x_i) in the data set we compute its range $[\min(x_i), \max(x_j)]$ and generate values for the n points uniformly from the interval $\min$ to $\max$.
- For the observed data and the reference data, the total intracluster variation is computed using different values of k .
- The gap statistic for a given k is defined as follow:

$$Gap_n(k) = E_n^*\{\log(W_k)\} - \log(W_k)$$

- Where E_n^* denotes the expectation under a sample of size n from the reference distribution.



- E_n^* is defined via bootstrapping by generating B copies of the reference datasets and, by computing the average $\log(W_k^*)$.
- The gap statistic measures the deviation of the observed W_k value from its expected value under the null hypothesis.



A motivation for the gap statistic is the following, consider clustering n uniform data points in p dimensions with k centers. In this case we can estimate a reference value for

$$E_n^*\{\log(W_k)\} = \log(pn/12) - (2/p)\log(k) + \text{Constant}.$$

- If data actually have K well-separated clusters, we expect $\log(W_k)$ to decrease faster than its expected rate $(2/p)\log(k)$ for $k \leq K$.
- When $k > K$, we are essentially adding an (unnecessary) cluster centre and then $\log(W_k)$ should decrease more slowly than its expected rate.
- Hence the gap statistic should be largest when $k = K$.



The standard deviation (sd_k) of $\log(W_k)$ is also computed in order to define the standard error (s_k) of the simulation as follow:

$$s_k = sd_k \times \sqrt{1 + 1/B}$$

because more robust approach is to choose the optimal number of clusters K as the smallest k such that

$$Gap(k) \geq Gap(k+1) - s_{k+1}$$

That is, we choose the smallest value of k such that the gap statistic is within one standard deviation of the gap at $k+1$:

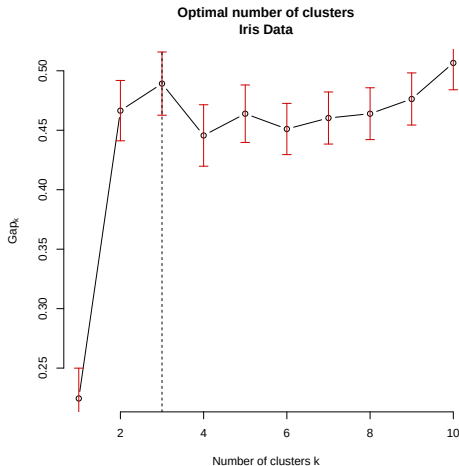
$$s_{k+1} \geq Gap(k+1) - Gap(k)$$



- 1 Cluster the observed data, varying the number of clusters from $k = 1, \dots, k_{max}$, and compute the corresponding W_k .
- 2 **Generate B reference data sets** and cluster each of them with varying number of clusters $k = 1, \dots, k_{max}$. Compute the estimated gap statistic $Gap_n(k) = E_n^*\{\log(W_{kb})\} - \log(W_k)$.
- 3 Let $\bar{w} = (1/B) \sum_b \log(W_{kb}^*)$, compute the standard deviation $sd(k) = \sqrt{(1/B) \sum_b (\log(W_{kb}^*) - \bar{w})^2}$ and define $s_k = sd_k \times \sqrt{1 + 1/B}$.
- 4 Choose the number of clusters as the smallest k such that $Gap(k) \geq Gap(k+1) - s_{k+1}$.

Very “time consuming”!





- 5 How Many Clusters?
- The Elbow Method
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But, wait, what is cluster?



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- The definition of a community or cluster, is not an easy task [EC02].
- Already in 1964, Bonner [Bon64] argued that there could not be a universal definition of a cluster. In fact, what is a cluster is in the eyes of the beholder, and one person's noise could be another person's signal [HK06].
- That is why we can write that cluster analysis is structure seeking although its operation is structure imposing [AB84].



- Even with a given single and clear definition of a cluster, we expect from an adapted clustering that it gives a good clustering, but, again, what is a good clustering?
- Knowing that clustering algorithms transform clusters's research into optimization problem whose computational complexity is typically intractable and which are solved by approximation algorithms [EC02].



Kleinberg has proved in the impossibility theorem [Kle03] that, given a distance function d there is no clustering method that satisfies the three following properties:

- scale-invariance: for any constant α the new distance $\alpha \cdot d$ must give the same clusters,
- richness: given a partition, we must be able to construct a distance function d which gives this partition,
- consistency: two distances d and d' which are such, $d'(i,j) \leq d(i,j)$ when i and j belong to the same cluster, and $d'(i,j) \geq d(i,j)$ otherwise, must give the same partition.



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